EE426 / 626 Microprocessor Applications

Brief History of the Intel 8048 (MCS-48) Chipset



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Intel 8048/49

MCS-48 Family

1976 Intel releases 8048

First Microcontroller (MCU)

3 μm HMOS (depletion-mode NMOS) → 2 μm

40-pin PDIP/CDIP

Modified Harvard Architecture

- 12-bit address (4 kB program ROM)
- 64 to 256 bytes on-chip data RAM

27 I/O Lines

8-bit Timer / Counter

Single +5 V_{CC} Supply with 6 MHz clock

Interrupt support with banked context registers

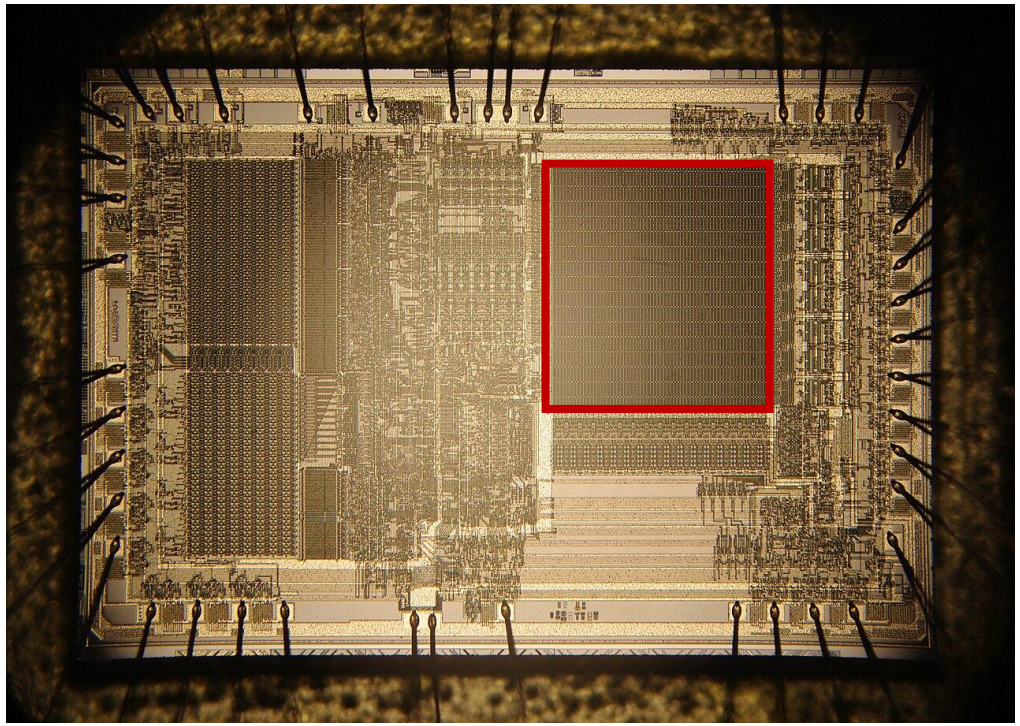
90+ Instructions



By Konstantin Lanzet - CPU collection Konstantin Lanzet, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=6589779

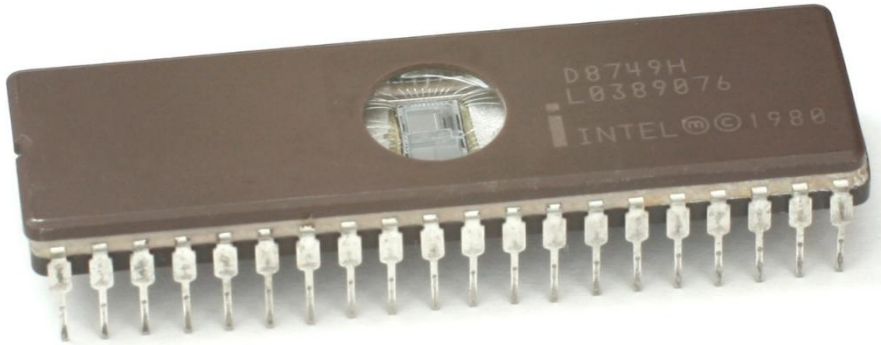
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Intel 8749H with 2 kB x 8 EPROM



By yellowcloud from Germany - EPROM-Microcontroller Intel 8749H Uploaded by Jacopo Werther, CC BY 2.0, https://commons.wikimedia.org/w/index.php?curid=24975860

Intel 8749H with 2 kB x 8 EPROM



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IBM PC 101-key Keyboard (1986)



By Raymangold22 - Own work, CC0, https://commons.wikimedia.org/w/index.php?curid=37410391

TRS-80 Model II (1979)



By 4throck - Own work,Okona, https://commons.wikimedia.org/wiki/File:Okona-GfhR-TRS-80.jpg, CC BY 2.5, https://commons.wikimedia.org/w/index.php?curid=169555494

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Roland Jupiter-4 Sythesizer



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Nintendo Donkey Kong





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MCS-48 family:



Intel P8048H

- 8020 – Single-Component 8-bit Microcontroller, 1 KB ROM, 64 Byte RAM, 13 I/O ports
- 8021 – Single-Component 8-bit Microcontroller, 1 KB ROM, 64 Byte RAM, 21 I/O ports
- 8022 – Single-Component 8-bit Microcontroller, 2 KB ROM, 64 Byte RAM with on-chip A/D Converter
- 8035 – 8-bit Microcontroller, 64 Byte RAM
- 8039 – 8-bit Microcontroller, 128 Byte RAM
- 8040 – 8-bit Microcontroller, 256 Byte RAM
- 8048 – Single-Component 8-bit Microcontroller, 1 KB ROM, 64 byte RAM, 27 I/O ports, 0.73 MIPS @ 11 MHz
- 8049 – Single-Component 8-bit Microcontroller, 2 KB ROM, 128 byte RAM, 27 I/O ports,
- 8050 – Single-Component 8-bit Microcontroller, 4 KB ROM, 256 byte RAM, 27 I/O ports,
- 8748 – Single-Component 8-bit Microcontroller, 1 KB EPROM, 64 byte RAM, 27 I/O ports,
- 8749 – Single-Component 8-bit Microcontroller, 2 KB EPROM, 128 byte RAM, 27 I/O ports,
- 87P50 – 8-bit Microcontroller, ext. ROM socket (2758/2716/2732), 256 byte RAM, 27 I/O ports
- 8648 – Single-Component 8-bit Microcontroller, 1 KB OTP EPROM, 64 byte RAM, 27 I/O ports
- 8041 – Universal Peripheral Interface 8-bit Slave Microcontroller, 1 KB ROM, 64 byte RAM
- 8041AH – Universal Peripheral Interface 8-bit Slave Microcontroller, 1 KB ROM, 128 byte RAM
- 8641 – Universal Peripheral Interface 8-bit Slave Microcontroller ?
- 8741 – Universal Peripheral Interface 8-bit Slave Microcontroller, 1 KB EPROM, 64 byte RAM
- 8741AH – Universal Peripheral Interface 8-bit Slave Microcontroller, 1 KB EPROM, 128 byte RAM
- 8042 – Universal Peripheral Interface 8-bit Slave Microcontroller, 2 KB ROM, 256 byte RAM
- 8742 – Universal Peripheral Interface 8-bit Slave Microcontroller, 2 KB EPROM, 128 byte RAM
- 8742AH – Universal Peripheral Interface 8-bit Slave Microcontroller, 2 KB OTP EPROM, 256 byte RAM